

2 ARCHITECTURE MAP II: WHERE COMPUTATION LIVES

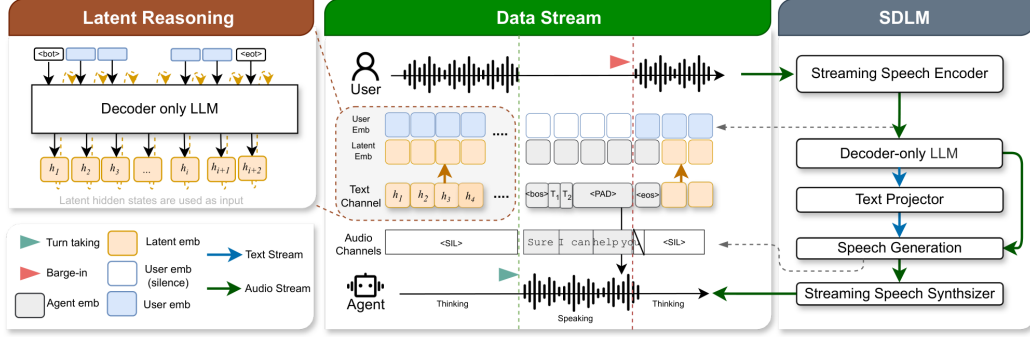


Figure 3: **FLAIR**. Reproduced from Wu et al. (2026), Fig. 2. Qwen2.5-7B: latent reasoning while listening; text while speaking. Full-context teacher is training-only.

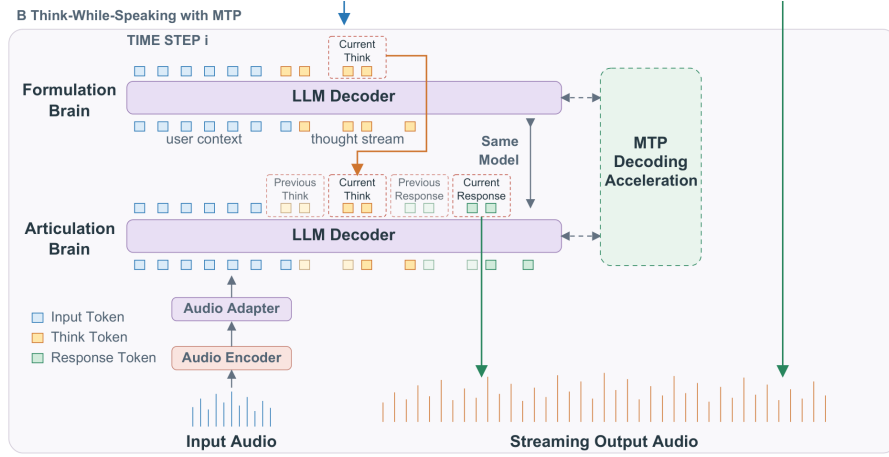


Figure 4: **StepAudio 3 Realtime**. Reproduced from Lin et al. (2026a), Fig. 7B (routing panel A omitted). Same-model formulation/articulation; playback-aware pacing follows Mind-Paced Speaking (Wu et al., 2025).

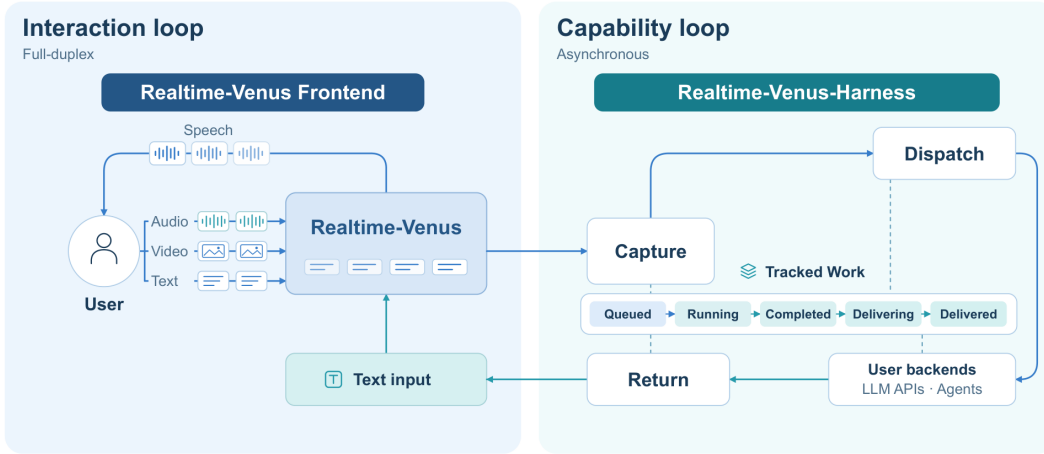


Figure 5: **Realtime-Venus**. Reproduced from Ant Group (2026), Fig. 3. Foreground interaction continues during asynchronous delegation; the frontend controls reply delivery timing.

3 SYNCHRONIZATION: INCOMING SPEECH VS. PLAYED SPEECH

The original proposal centers on *prefix sufficiency*, *correction-aware reasoning* and *speaker ownership*. **Listening sync:** track how much of each speaker’s speech the model has *actually processed*, not just received. **Speaking sync:** distinguish generated speech, buffered audio and speech *already played*. Autoregressive steps are not wall-clock seconds. Wrong claims already played need a spoken correction; buffered audio can still be changed.

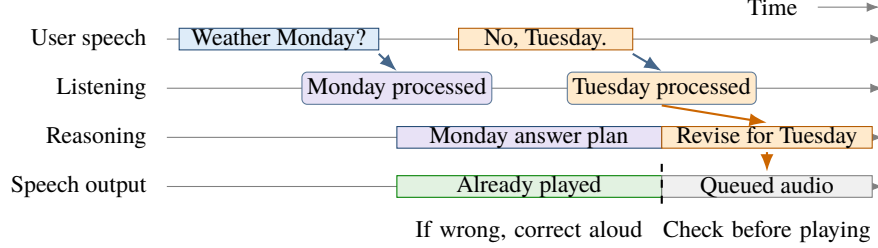


Figure 6: **Illustrative example, not an experiment.** User speech enters as streaming audio tokens or features, depending on the model. Once the Tuesday correction is processed, revise the answer plan and check buffered audio before playback. A wrong claim already played cannot be taken back; correct it aloud.

Table 1: Five closest synchronization references: what already exists.

Paper	Established mechanism
SynclLM (Veluri et al., 2024)	Clock-aligned chunks; speculative user tokens replaced by observations.
Moshi (Défossez et al., 2024)	Parallel user/agent codec streams on an 80 ms frame grid.
DuplexSLA (Zhang et al., 2026)	160 ms speech/listening/action clock; release marked forthcoming.
AdaptDuplex (Zhou et al., 2026)	Adaptive windows, bounded text lead, cancel / stale-result suppression.
Synchronization / turn-taking (Riera et al., 2026)	Moshi–Moshi hidden-state coupling and causal timing probes, not reasoning correctness.

Table 2: Five discussion questions: small falsifiable tests, not claimed results.

Question	First paired test	Measure
RQ1: Think while listening When is a prefix sufficient?	Late constraints with identical prefixes; risk-calibrated vs. fixed triggers.	Correct useful-answer time; premature claims.
RQ2: Think while speaking How much speech lead is safe?	Corrections before / after playback; bounded vs. unrestricted queue.	Stale audible seconds; repair; underruns.
RQ3: Listen while speaking Whose correction updates what?	Relevant correction vs. backchannel / side speech; oracle vs. predicted identity.	Correction accuracy; false interruption; silence.
RQ4: Synchronize the clocks Which schedule survives load?	Same audio under rate / jitter / contention; fixed vs. adaptive quantum.	p95 evidence age; useful-answer delay; quality.
RQ5: Revise while delegating Which returned answer is valid?	Correction during a simulated backend job; cancellation vs. joint result / playback validation.	Stale-result use; task success; wasted work.

Novelty boundary. Early thought, dual-process articulation, adaptive windows and stale-job suppression already exist (Shih et al., 2026; Wu et al., 2025; Zhou et al., 2026). The candidate contribution is *joint speaker/evidence/playback consistency*, tested against those mechanisms. Five questions are discussion options, not ranked winners.

4 RESOURCE SHEET AND A SMALL FIRST TEST

Table 3: Five evaluation choices: complementary, not one leaderboard.

Benchmark	What it tests	Access / interpretation
SRQA (Shih et al., 2026)	Streaming reasoning accuracy vs. post-question CoT delay.	Synthetic; recipe verified, released evaluation not located. CoT onset is not a useful answer.
τ -Voice (Ray et al., 2026)	Grounded tool/task success across 278 tasks.	Code / tasks / simulator; APIs needed. Virtual time is not physical deadline validation.
Full-Duplex-Bench v3 (Lin et al., 2026b)	Human disfluencies, tools, argument accuracy and completion.	Code + recordings. Tool F1 does not equal task success.
EchoChain (Modi et al., 2026)	State revision during speech: inertia, amnesia, goal loss.	200 synthetic interrupted conversations; standalone release not located.
Duplex-MPE (Ma et al., 2026)	Addressee control, silence, answers and floor release.	4,000 synthetic streams; standalone release not located.

Table 4: Five supervision sources: natural timing is not a reasoning label.

Resource	Useful content / scale	Access / caution
SmoothConv (QualiaLabs, n.d.)	100.53 h Chinese; speaker / turn-state / timing labels.	HF; CC BY-NC 4.0; shared sources with DuplexConv.
DuplexConv (OpenT2S, n.d.)	2,000.21 h Chinese; channels, VAD and machine-assisted labels.	HF; CC BY-NC 4.0; split by source session.
otoSpeech Task (otoEarth, n.d.-a)	20.006 h / 58 English sessions; channels and task events.	Gated; CC BY 4.0; verify task-outcome labels.
otoSpeech conversational (otoEarth, n.d.-b)	280 h raw / 141 h curated English; separate speaker channels.	Gated; CC BY 4.0; curated data is a subset.
Open Yap 1K (The Agentic Data Company, n.d.)	1,000 h English; word timing, familiar-speaker dyads.	Public sample; full corpus requires request / DUA.

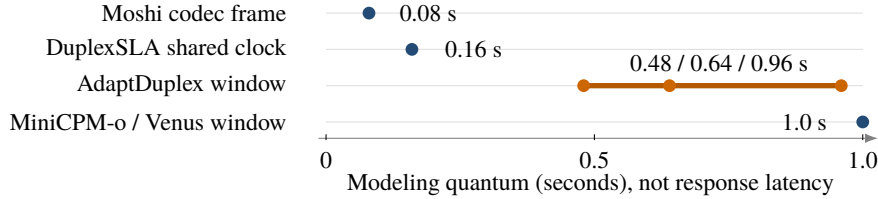


Figure 7: **Published timing parameters, not a latency leaderboard.** Codec frames, shared action clocks and language windows are different quantities (Défossez et al., 2024; Zhang et al., 2026; Zhou et al., 2026; Cui et al., 2026; Ant Group, 2026). Smaller quantum does not imply a faster *correct* answer.

First test. Instrument a released 7–9B runtime on 200–500 paired dialogues: relevant correction vs. backchannel; before vs. after playback; same vs. different speaker. Keep model, lookahead and compute budget fixed; use a simulated read-only backend. Log ingestion, state version, output queue and playback. Compare *correct useful-answer latency*, *stale audible seconds*, *final task success* and *false interruption*, not first filler sound. Natural corpora need added sufficiency/revision labels; hold out benchmark tests. No experiments or performance measurements were run for this brief.

Literature/availability cutoff: 30 September 2026. “Not located” is not proof of nonexistence. Details and caveats: see the preserved nine-page report and source notes.

5 REPORTED PERFORMANCE ON THE SELECTED BENCHMARKS

These are source-reported results, not our experiments. Original benchmark comparisons and later system-report evaluations are labeled separately; their model versions, conditions and aggregation can differ.

Table 5: Reported performance (percent, higher is better). Scores follow the metric order in each row; compare only within a protocol. NR: selected-system result not located, not zero. Closed references are the best in the stated source comparison, not a current global leaderboard.

Benchmark / metric order	Selected systems	Best closed voice in source
SRQA ICLR proceedings; ARC-E / ARC-C / SIQA / PIQA / GSM8K (Shih et al., 2026)	Moshi + CoT (TWL study) 77.7 / 59.8 / 56.1 / 56.9 / 16.1 TWL: early length-DPO 65.4 / 46.0 / 45.3 / 46.0 / 14.7	No closed voice result.
τ-Voice Original benchmark, Table 6 All row; clean / realistic (Ray et al., 2026)	NR	Grok Voice 51 / 38
τ-Voice (AA) Artificial Analysis implementation; equal-domain macro task-success rate (Lin et al., 2026a)	StepAudio 3 Realtime 56.0	Grok Voice Think Fast 2.0 High 56.5
Full-Duplex-Bench v3 Tool selection F1 / argument accuracy / pass@1 (Lin et al., 2026b; Ant Group, 2026)	Realtime-Venus-Omni 86.0 / 53.1 / 43.0 Realtime-Venus-Audio 82.0 / 52.2 / 42.0	GPT-Realtime 87.6 / 68.0 / 60.0
EchoChain 200 interrupted conversations; conversation MPR / criterion MCP (Modi et al., 2026)	NR	Grok Voice Agent 48.5 / 85.5
Duplex-MPE Explicit / implicit addressing; fresh onset / conditional answer / silence / yield (Ma et al., 2026)	MiniCPM-o 4.5 (explicit) 95.00 / 47.30 / 92.42 / 59.64 MiniCPM-o 4.5 (implicit) 94.35 / 46.65 / 92.89 / 59.25	No closed voice result.

Text controls, not voice competitors. Original τ -Voice: GPT-5 reasoning, 85% task pass@1. Duplex-MPE: Gemini 3.1 Pro with speaker-attributed transcripts, explicit 94.70 / 90.39 / 95.37, implicit 30.40 / 79.11 / 99.34 (response presence / conditional answer / silence, %; Table 17 reference run). No fresh-onset or stopping score; not a speech-model upper bound. (Ray et al., 2026; Ma et al., 2026)

Coverage and configuration caveats. FLAIR has no located score on these five benchmarks; its Full-Duplex-Bench results use an earlier version. SRQA’s CoT and early-DPO rows are different checkpoints (Tables 2 and 4); early-DPO uses question-completeness threshold 0.75, with delays of 12.0 / 13.2 / 12.9 / 18.2 / 48.6 tokens in the same task order, not runtime seconds. Venus’s v3 scores come from its revised report, not a common rerun with the benchmark authors. The two τ -Voice rows are not interchangeable. Duplex-MPE yield has model-specific eligibility; Gemini’s text reference supplies speaker identity. Unlisted selected-system/benchmark pairs are NR.

Reading the gaps. Tool selection can be strong while complete-task success remains weak; a response can occur without being correct or appropriate. The missing cross-system coverage is itself a reason to build a shared, version-pinned evaluation harness, not evidence that an untested system fails.

AVAILABILITY AND REFERENCES

Table 6: Five system anchors: availability snapshot, 30 September 2026.

System	Inference / weights	Training reproducibility
Think while listening	Study release not located; base Moshi is released.	Recipe; full spoken-CoT data not located.
FLAIR	Dedicated code / weights not located.	Latent ELBO/SFT; full mixture not located.
MiniCPM-o 4.5	Released code + weights; Apache-2.0.	Fine-tuning tools; complete duplex mixture not released.
StepAudio 3 Realtime	Report / demos; Realtime weights not located.	Methods; full code / mixture not located.
Realtime-Venus	Released harness + Audio/Omni weights; Apache-2.0.	Methods; complete training package not located.

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